arc-fs: adaptive resampling with pruned trees

The AdaBoost of Breiman's comparisons, tested against his published test errors on 20 data sets.

Tested against	Breiman (2001), *Random forests*, Table 2, the AdaBoost column (p. 11) and Section 6 (p. 15); Breiman (1998), *Arcing classifiers*, Tables 3 and 5. 15 real data sets from PollisDatasets, four simulated as Breiman did, and the zip-code digits of Hastie, Tibshirani and Friedman (2009) (downloaded only if the user agrees).
Missing in Julia	arc-fs, and CART pruning by cost complexity on a test sample. DecisionTree.jl grows the trees only.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *arc-fs: adaptive resampling with pruned trees* (`arcfs.ipynb`)
Result	18 of 20 test errors within simulation noise of the published ones.

The problem

arc-fs is the version of AdaBoost of Freund and Schapire (1996) behind Breiman's comparisons: the "AdaBoost" column against which random forests were first judged. Each step draws a training set that favours the cases earlier trees got wrong, grows a CART tree, prunes it, and the trees vote with weights. No Julia package has arc-fs, and none prunes a CART tree by cost complexity on a test sample. Without both, the classic comparison of boosting and forests cannot be rerun in Julia.

The method

Cost-complexity pruning

CART, Breiman, Friedman, Olshen and Stone (1984), Ch. 3, repeatedly collapses the weakest link, the node t with the smallest $g(t) = (R(t) - R(T_t))/(|T_t| - 1)$, where $R(t)$ counts the cases misclassified at t as a leaf, $R(T_t)$ those misclassified by the branch below it and $|T_t|$ its leaves. `prune` keeps, from this sequence, the subtree with fewest errors on a second sample, the smallest on ties.

arc-fs

A probability $p(n)$ sits on each training case, equal at the start. At step k, draw a training set with these probabilities, grow a tree and prune it on a second draw. With $d(n) = 1$ for the cases the tree misclassifies,

$$\varepsilon_k = \sum p(n)\, d(n), \qquad \beta_k = (1 - \varepsilon_k)/\varepsilon_k, \qquad p(n) \leftarrow p(n)\, \beta_k^{d(n)}, \text{ normalised,}$$

and the trees vote with weights $\log \beta_k$. When ε_k reaches $\frac{1}{2}$, or is 0, the probabilities are reset and the construction restarts; that step's tree does not vote.

What the tests show

On each small data set a random 10% is the test set and 50 trees are combined; 100 runs are averaged. Letters, sat-images and zip-code have their own test sets and are one run each, with a binomial standard error. A value is *within noise* when the two differ by less than two standard errors of a difference ($\sqrt{2}$ s.e.).

Data sets	What is compared	Within noise
13 small real sets	average test error of 100 runs	13 of 13
letters, sat-images, zip-code	one run on their own test sets	3 of 3
twonorm, threenorm, ringnorm, waveform	average test error of 50 runs	2 of 4
All	20 test errors	18 of 20

Where they differ

Twonorm 4.6 vs 4.9 and ringnorm 6.2 vs 6.9: below Breiman by 2.8 and 4.4 standard errors. Among those that agree, breast cancer (3.7 vs 3.2) and diabetes (25.3 vs 26.6) came closest to the limit. The notebook's example shows the point of arcing: on one split of the breast cancer data, the vote of a few dozen pruned trees halves the error of a single one.

Still open

- **Twonorm and ringnorm.** The random forests of the Out-of-bag notebook are below Breiman on the same two sets. A gap shared by two methods points to how the trees are grown, or to the generators, rather than to arc-fs. Not explained.

- **Details the papers leave open:** whether a restarting step's tree joins the vote (here it does not), and how CART chose among equally good pruned subtrees (here the smallest).
- **The data.** Soybean has 683 cases, not 685, and its categorical inputs are split as ordered codes, since DecisionTree.jl has no splits on subsets of categories. Missing values are replaced by medians; Breiman does not say what he did.

Further considerations

- **Pruning on its own.** Cost-complexity pruning is useful well beyond arc-fs: it is how a single CART tree is usually chosen, and it is missing from DecisionTree.jl. It could be offered upstream with these tests.
- **Categorical splits.** Splits on subsets of categories, which Breiman's CART had, would remove the soybean caveat and are a natural next contribution.
- **Next.** The arcing paper's arc-x4 and bagging columns, from the same tables, to rerun the whole comparison.

References

Breiman, L. (1998). Arcing classifiers. *The Annals of Statistics* 26(3), 801–849. doi:10.1214/aos/1024691079

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Breiman, L., Friedman, J. H., Olshen, R. A. and Stone, C. J. (1984). *Classification and Regression Trees*. Wadsworth.

Freund, Y. and Schapire, R. E. (1996). Experiments with a new boosting algorithm. *Proceedings of the Thirteenth International Conference on Machine Learning*, 148–156.

Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer: the zip-code data.

Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.
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